

#12 Inventory Slack Alert System

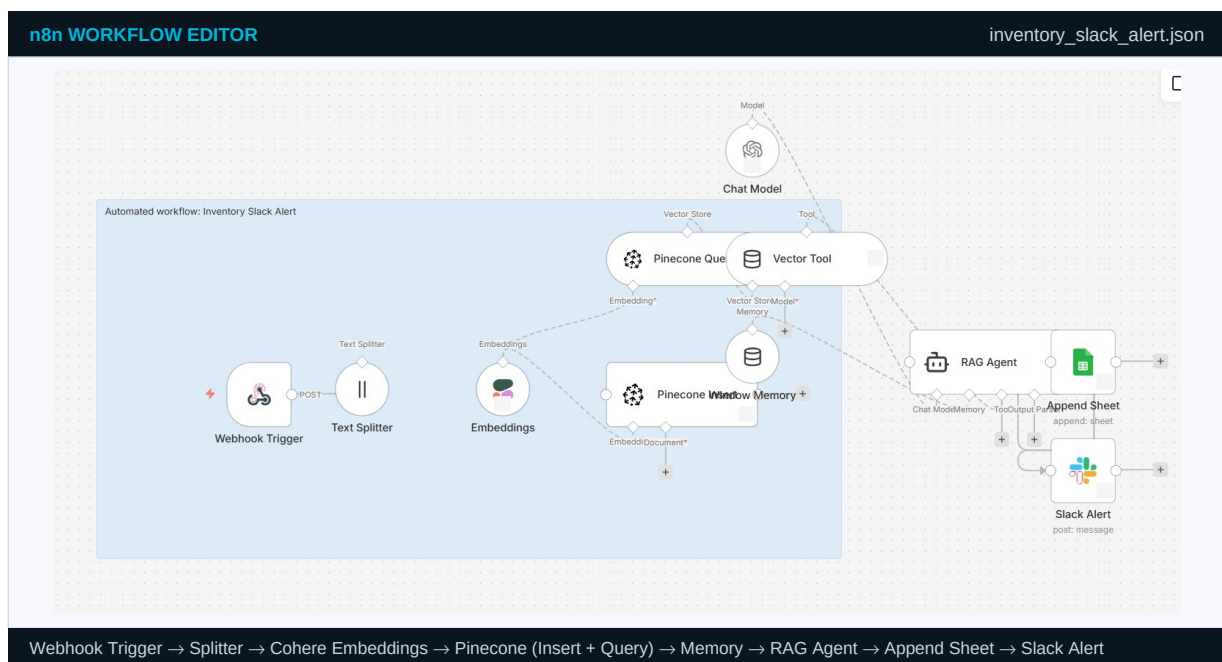
E-Commerce /
Inventory

Never stock out — AI calculates days-to-zero and alerts before the crisis hits

BUSINESS PROBLEM

Stock-outs cost e-commerce stores around 4% of annual revenue. This workflow monitors every SKU continuously, computes real days-to-stockout from actual sales velocity, and fires tiered Slack alerts with supplier contacts and recommended reorder quantities — catching problems 2 weeks early.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — ERP inventory sync (every 30 min)

```
{
  "sku": "NK-RS-X200-BLK-10",
  "product_name": "Running Shoes X200 Black Size 10",
  "current_stock": 4,
  "warehouse": "Warehouse-A",
  "last_sale_ts": "2024-06-15T07:44:00Z",
  "reorder_point": 20,
  "site": "main-store"
}
```

OUTPUT — Slack alert — tiered stock warning with reorder details

```
{
  "json": {
    "sku": "NK-RS-X200-BLK-10",
    "current_stock": 4,
    "daily_rate": 2.1,
    "days_to_zero": 1.9,
    "alert_level": "RED",
    "reorder_qty": 120,
    "supplier": "Nike Distribution Ltd",
    "lead_time_days": 14,
    "supplier_phone": "+44-161-XXX-XXXX",
    "estimated_cost": "GBP 4,200",
    "slack_sent": true,
    "ticket_created": true
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — ERP inventory sync	Every 30 min: {sku, product_name, current_stock, warehouse, last_sale_timestamp}. Triggered on every sync event.
2	➤ Splitter — product catalogue chunks	Supplier details, lead times, MOQs from product catalogue embedded into Pinecone for agent lookup.
3	■ Cohere Embeddings — product knowledge	'Nike supplier: 14-day lead time, MOQ 50 units, contact: +44-161-XXX' embedded as retrievable knowledge.
4	■ Pinecone Insert — stock snapshot	Each reading stored with timestamp. Enables: $\text{daily_rate} = (\text{prev_stock} - \text{curr_stock}) / \text{hours elapsed}$.
5	■ Pinecone Query — 30-day sales history	Retrieves 30-day stock levels per SKU. Agent computes daily_sales_rate and trend_direction (accelerating/stable).
6	■ Vector Tool — supplier lookup	Agent fetches: supplier name, lead_time_days, MOQ, last_order_date, emergency_contact for this SKU.
7	■ Window Memory — 10 readings	Tracks short-term changes: sudden spike = promotion running. Steady drain = normal sales velocity.
8	■ RAG Agent — risk calculation and alert formatting	$\text{days_to_stockout} = \text{current_stock} / \text{daily_rate}$. $\text{Reorder_qty} = (\text{lead_time} * \text{daily_rate}) * 1.3 \text{ safety factor}$.
9	■ Append Sheet — inventory log	All alerts: sku, stock_level, days_to_zero, alert_level, reorder_qty, supplier. Weekly SKU risk report.
10	■ Slack — tiered notification	RED (<5 days) → #urgent-ops with @channel. AMBER (<14 days) → #inventory-mgmt. Includes supplier phone.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
Cohere API Key	dashboard.cohere.com → API Keys	Cohere API
Pinecone API Key	app.pinecone.io → API Keys	Pinecone API
OpenAI API Key	platform.openai.com	OpenAI API
Slack Bot Token	api.slack.com/apps → OAuth & Permissions	Slack API

SAMPLE RUN — what this workflow processed and produced

Scenario	An online footwear retailer stocks 840 active SKUs. The buying team previously did manual stock checks weekly — by which time several SKUs had already stocked out and orders were being cancelled.
Input received	At 08:00 sync: Running Shoes X200 Black Size 10 showed 4 units remaining. Previous sync 30 minutes earlier had shown 5 units. Daily rate computed at 2.1 units/day.
What the workflow found	$4 \text{ units} / 2.1 \text{ per day} = 1.9 \text{ days to stockout}$. Nike lead time: 14 days. Reorder needed immediately. This SKU had been in AMBER status for 6 days without action — now escalated to RED.

Output produced	RED alert posted to #urgent-ops at 08:01 with supplier phone number. Buyer created PO for 120 units (MOQ 50, safety factor applied) at 08:23. Stockout avoided. Next shipment due June 29.	
JSON: inventory_slack_alert.json	GitHub: trm-inventory-alert-agent	www.shopify.com/blog/inventory-management